Fiscal Year:	FY 2007	**Task Last Updated:**	FY 09/25/2007
PI Name:	Brady, Joseph V. Ph.D.		
Project Title:	Psychosocial Performance Factors in Space Dwelling Groups		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	21224-6823	**Congressional District:**	3
Comments:	NOTE: PI Dr. Joseph Brady passed away in July 2011.		
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	04/01/2004	**End Date:**	04/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 4/30/2008 per K. Major/NSBRI (3/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hienz, Robert (The Johns Hopkins University School of Medicine) Hursh, Steven (Institutes for Behavior Resources, Inc.)		
Grant/Contract No.:	NCC 9-58-NBPF00403		
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Performance Goal Text:			

Task Description:	The original aims of this project focused upon the development of an experimental test bed for modeling performance effectiveness and psychosocial adaptation in support of exploratory missions beyond Earth's atmosphere. Despite uncertainties regarding the performance requirements of projected space laboratories, workstations, Lunar modules, interplanetary vehicles, and space settlements, a common feature of such spaceflight endeavors over the next half century will be extended stays by human groups in extraterrestrial habitats. The research methodology adopted to accomplish these objectives provided for distributed interactive multi-person simulations in computer-generated environments focused upon psychosocial performances within and between space-dwelling flight crews and Earth-based control center groups. The experimental setting provided the automated means for setting the context for the analysis of space crew performance as well as monitoring electronically the interactive effects of simulated communication modality constraints and other environmentally stressful conditions. Distributed interactive simulation experiments characterized the extent to which variations in the structure and function of communication channels within and between space-dwelling and Earth-based groups affect the ability to accurately discern and respond to the realities and requirements of mission operations. The effects of environmentally stressful time constraints as well as the appetitive and aversive characteristics of programmed incentive conditions have been analyzed in the context of spaceflight simulations. The initial results of these distributed interactive simulation studies requiring the identification, collection and analysis of geologic specimens designated by a range of complex rules showed clearly that cooperative and productive interactions were maintained between individually isolated and dispersed crewmembers in a task-driven environment. All experimental flight crews actively engaged in communicating and effective problem-solving over extended time intervals without benefit of one another's physical presence. In addition, the investigations of communication modality constraints indicated that there was a high degree of interchangeability between available communication modes as well as use of 'indexical' features of the simulation display as countermeasures to maintain effective crew performance. In extending these studies to the analysis of crew configuration changes in simulated space-dwelling groups, crewmates were reassigned to simulated vehicles other than those normally operated. These reassignments based upon baseline performance and variations in crewmate skill levels resulted in "Strong" and "Weak" crew configurations. An embedded measure of efficiency was developed based upon the "lag time" between the receipt of geologic specimen collection data and submitting the specimen for analysis indicating the extent to which specimen data was processed in a timely manner. Replications of configuration change experiments with several simulated spaceflight crews showed clearly that baseline performance effectiveness levels under experienced "Strong" crew configuration conditions were significantly reduced during missions following crew assignment changes that resulted in "Weak" crew configurations. Analysis of "lag time" data provided an effective account of the observed performance changes. Crew composition variations were also shown to have effects upon group dynamics that were not necessarily reflected by the crew performance measures. When such adverse psychosocial side effects did occur however, they were not long-lasting and all self-report measures (Visual Analogue Scale, Mission Debriefing Reports) returned to baseline levels when the original crew configuration was reinstated. Studies of incentive effects were initiated under conditions that made monetary rewards contingent upon simulation scenario performance effectiveness. The task performances of individual crewmembers contributed to a group bank account with the proceeds divided equally among the crew members upon completion of the experiment. Under positive incentive conditions, crew performances that exceeded a predetermined geological specimen grade value standard resulted in proportional additions to the group bank account. Under negative incentive conditions, crew performances that were below a predetermined grade value standard resulted in proportional subtractions from the group bank account. A Visual Analogue Scale (VAS) administered by computer to crewmembers for each experimental session displayed positive and negative terms (e.g. "happy", "frustrated") and provided quantitative measures of behavioral and psychosocial adaptation.. Crew performance effectiveness increased significantly under both positive and negative incentive conditions. In contrast, the self-report VAS measures reflected a consistent change in psychosocial adaptation with negative incentive conditions characterized by a pronounced increase in negative self-report ratings accompanied by decreased positive self-report ratings. Positive incentive conditions were characterized by positive self-report ratings only. The results of these experiments show clearly that incentive conditions associated with simulated spaceflight missions can significantly affect psychosocial adaptation without compromising task performance effectiveness in trained and experienced volunteer crewmember participants. Future plans will involve migration of the simulation scenario providing for a broader range of Lunar surface regions to enhance the complexity of geologic specimen identification, collection and conceptual analysis task. Variations in core communication functions under this more complex model will focus upon an experimental analysis of the interactions between ground-based control and flight crew performance effectiveness.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Research conducted within the context of distributed interactive simulation models will provide the basis for developing effective patterns of communication and problem solving strategies as well as a range training procedures to enhance problem solving effectiveness. The Earth-based applications of this research will extend to the small operational group selection and training process as well as the management of stressful interactions and the maintenance of group cohesion and productivity. Not only can the outcome of these studies be expected to have an important impact on safety and the quality of life in many Earth-based applied settings, but larger societal units will ultimately benefit from the resulting conceptual and methodological advances that effectively promote social and ecological stability while concurrently enhancing an education and training technology that assures effective communication of an expanded generalizeable knowledge base.
	Task progress included an extended analysis of the crew configuration data completed during the current reporting year confirmed that the changes in the embedded lag time key problem solving measure associated with variations in crew assignments were not reflected in other performance measures (e.g. communication frequencies, specimen collection rates, etc.) and were not affected by incentive conditions. Although crew composition changes were also accompanied by alterations in psychosocial adaptation, such adverse side effects were not long-lasting and all self-report measures (Visual Analogue Scale, Post-Mission Debriefing Reports) returned to baseline levels when the original crew configuration was reinstated. Similarly transient adverse psychosocial side-effects were as well observed in response to introduction of the incentive conditions even though they had no apparent effect on performance effectiveness levels reflected in the embedded lag time measure.

Task Progress:	The major task progress during the current reporting period however, has been associated with an experimental analysis of incentive conditions and their relationship to the behavioral and psychosocial adaptation factors that effect both individual and crew performance effectiveness. The procedure developed provided for differential incentives contingent upon task performance levels on the simulation geologic specimen scenario. Task performances by individual crewmembers contributed to a group bank account with the proceeds distributed equally among the group upon completion of an experimental series. Under positive incentive conditions, crew performances that exceeded a predetermined group standard resulted in proportional additions to the group bank account. Under aversive or negative incentive conditions, crew performances that fell below a predetermined group standard resulted in proportional subtractions from the group bank account. Crew performance effectiveness increased significantly under both positive and negative incentive conditions. In contrast, the self-report Visual Analogue Scale measures reflected a consistent change in crewmember psychosocial adaptation under negative incentive conditions. Such negative incentive conditions were associated with pronounced increases in negative self-report measures (e.g. “frustrated”) and concurrently decreased positive self-report ratings (e.g. “happy”). Positive incentive conditions were in contrast, associated with increased positive self-report ratings only, showing clearly that such incentive conditions can significantly affect psychosocial adaptation without compromising task performance effectiveness in trained and experienced crewmember volunteer participants.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Articles in Peer-reviewed Journals	Hienz RD, Brady JV, Hursh SR, Banner MJ, Gasior ED, Spence KR. "Communication constraints, indexical countermeasures, and crew configuration effects in simulated space-dwelling groups." Acta Astronaut. 2007 Feb-Apr;60(4-7):362-78. http://dx.doi.org/10.1016/j.actaastro.2006.09.005 , Feb-2007
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Awards	Brady JV. "Lawrence Young Award for Space Biomedical Research, February 2005." Feb-2005
Awards	Brady, JV. "Lifetime Research Contributions, Association for Behavior Analysis, Atlanta, Ga., May 2006." May-2006
Papers from Meeting Proceedings	Gazior ED, Spence KR, Hienz RD, Hursh SR, Brady JV. "Distributed interactive simulation in space dwelling groups." Huntsville Simulation Conference, October 2006. Proceedings, Huntsville Simulation Conference, 15 p., 2006. , Oct-2006